

Their recent US patent application describes a system in which an operator wears a head-mounted display and an exoskeleton suit while doing his tasks. Data from the suit and other external sensors is analysed by computer systems and used to control distant robots.

The wearable robotic suit includes head and neck motion sensors, devices to capture arm movements and haptic gloves. The operator can control the robot's movement using a foot pedal, and see what the robot is seeing using a virtual reality headset. The suit might also contain sensors and devices to capture brain waves.

The robot is described as a humanoid of 1.2-metre height, possibly covered with synthetic skin, with two (or more) arms ending in hands or grippers and wheeled treads for locomotion. It has cameras on its head and other sensors like infrared and ultra-violet imaging, GPS, touch, proximity, strain sensors and radiation detectors to stream data to its operator.

Something that catches everybody's attention here is a line that says, "An operator may include a non-human animal such as a monkey... and the operator interface may be... re-sized to account for the differences between a human operator and a monkey operator."

But what is so smart about an operator controlling a robot, even if the operator is a monkey? Well, the interesting part of this technology is that the robots will also, eventually, be able to learn from their operators and carry out the tasks autonomously. According to the patent application, device-control instructions and environment sensor information generated over multiple runs may be used to derive autonomous control information, which may be used to facilitate autonomous behaviour in an autonomous device.

Kindred hopes to do these using deep hierarchical learning algorithms like a conditional deep belief network or a conditional restricted Boltzmann

machine, a type of powerful recurrent neural network.

This is what possibly links Kindred to D-Wave. The operation of D-Wave's quantum computing system is described by these as being analogous to a restricted Boltzmann machine, and its research team is working to exploit the parallels between these architectures to substantially accelerate learning in deep, hierarchical neural networks.

In 2010, Rose also published a paper that shows how a quantum computer can be very effective at machine learning. So if Kindred succeeds in putting two and two together, we can look forward to a new wave of quantum computing in robotics.

Robots get more social

It is a well-known fact that technology can help disabled and vulnerable people to lead more comfortable lives. It can assist them to do their tasks independently without requiring another human being to help them. This improves their self-esteem.

However, Maja Mataric of University of South California, USA, believes that the technology can be more assistive if it is embodied in the form of a robot rather than a tool running on a mobile device or an invisible technology embedded somewhere in the walls or beds.

Mataric's research has shown that the presence of human-like robots is more effective in getting people to do things, be it getting senior citizens to exercise or encouraging autistic children to interact with their peers. "The social component is the only thing that reliably makes people change behaviour. It makes people lose weight, recover faster and so on. It is possible that screens are actually making us less social. So that is where robotics can make a difference—this fundamental embodiment," Mataric mentioned while addressing a gathering at American Association for the Advancement of Science.



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Email : sales@meconinst.com
Web : www.meconinst.com